

SYDNEY GRAMMAR SCHOOL



2015 Annual Examination

FORM V

MATHEMATICS EXTENSION 1

Monday 31st August 2015

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 100 Marks

- All questions may be attempted.

Section I — 9 Marks

- Questions 1–9 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II — 91 Marks

- Questions 10–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your name, class and Master on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your name, class and Master on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Ten.

5A: DS

5B: RCF

5C: SO

5D: DNW

5E: DWH

5F: REJ

5G: SJE

5H: KWM

Checklist

- SGS booklets — 7 per boy
- Multiple choice answer sheet
- Candidature — 124 boys

Examiner

DS

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

The trapezoidal rule is used to approximate a definite integral. If $n + 1$ function values are used, then we are summing the areas of how many trapezia?

- (A) $n - 1$
- (B) n
- (C) $n + 1$
- (D) $n + 2$

QUESTION TWO

What is the gradient of the curve $y = -e^{-x}$ at its y -intercept?

- (A) e
- (B) $-e$
- (C) 1
- (D) -1

QUESTION THREE

What is the equation of the horizontal asymptote of the curve $y = \frac{x - 2}{x - 3}$?

- (A) $y = 1$
- (B) $y = \frac{2}{3}$
- (C) $x = 3$
- (D) $x = 2$

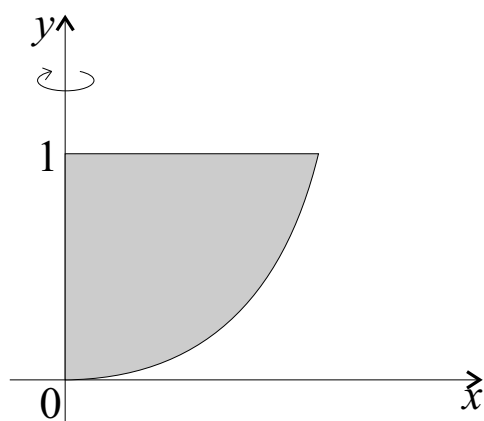
QUESTION FOUR

The definite integral $I = \int_{-2}^2 \sqrt{4 - x^2} dx$ can be evaluated without finding a primitive of $\sqrt{4 - x^2}$. What is the exact value of I ?

- (A) 4π
- (B) 2π
- (C) π
- (D) $\frac{\pi}{2}$

Exam continues next page ...

QUESTION FIVE



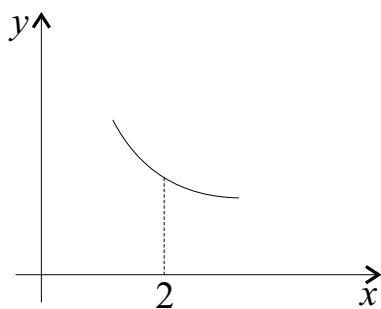
The diagram above shows the region bounded by the parabola $y = x^2$, the line $y = 1$ and the y -axis. What is the volume of the paraboloid formed by rotating this region about the y -axis?

- (A) $\frac{\pi}{5}$
- (B) $\frac{\pi}{3}$
- (C) $\frac{\pi}{2}$
- (D) $\frac{2\pi}{3}$

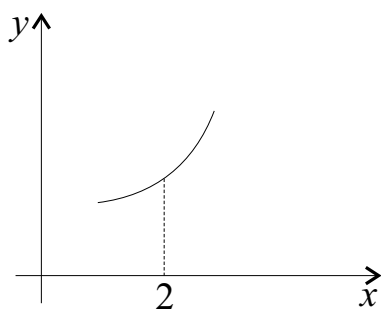
QUESTION SIX

A curve has equation $y = f(x)$. If $f'(2) < 0$ and $f''(2) > 0$, which diagram below shows the curve as it passes through the point where $x = 2$?

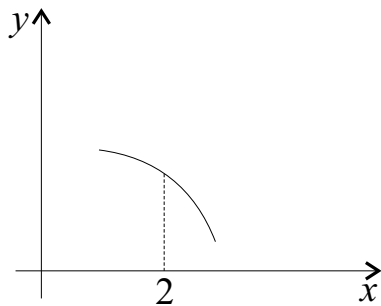
(A)



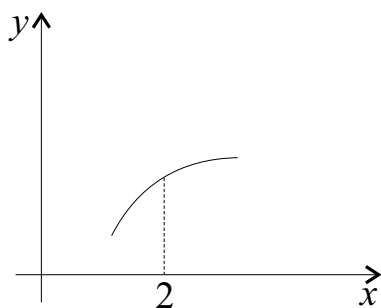
(B)



(C)



(D)



QUESTION SEVEN

By the chain rule, the derivative of $(x^2 + 1)^3$ is $6x(x^2 + 1)^2$. Which function is a primitive of $12x(x^2 + 1)^2$?

(A) $2x(x^2 + 1)^3$

(B) $2(x^2 + 1)^3$

(C) $\frac{1}{2}x(x^2 + 1)^3$

(D) $\frac{1}{2}(x^2 + 1)^3$

QUESTION EIGHT

Which expression is equivalent to $\frac{\sin \theta}{1 + \cos \theta}$?

(A) $\operatorname{cosec} \theta + \cot \theta$

(B) $\frac{1 - \sin \theta}{\cos \theta}$

(C) $\sin \theta + \tan \theta$

(D) $\frac{1 - \cos \theta}{\sin \theta}$

QUESTION NINE

For $x > 0$, which expression is NOT equivalent to $e^{\ln x}$?

(A) $\ln(e^x)$

(B) $x^{\ln e}$

(C) $(\ln e)^x$

(D) $\frac{1}{e^{\ln \frac{1}{x}}}$

————— End of Section I —————

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION TEN (13 marks) Use a separate writing booklet. **Marks**

(a) Differentiate:

(i) $(5 - 2x)^4$

1

(ii) xe^{2x}

2

(iii) $\log_e \sqrt{x}$

2

(b) Find:

(i) $\int \frac{1}{\sqrt{x}} dx$

1

(ii) $\int \frac{2}{3x + 4} dx$

1

(iii) $\int \frac{2}{(3x + 4)^2} dx$

2

(c) Evaluate $\int_e^{e^3} \frac{1}{x} dx$.

2

(d) The function $f(x)$ is defined by:

2

$$f(x) = \begin{cases} kx & \text{for } x < 2 \\ x^2 + 6 & \text{for } x \geq 2 \end{cases}$$

For what value of k is $f(x)$ continuous at $x = 2$?

QUESTION ELEVEN (13 marks) Use a separate writing booklet.**Marks**

- (a) Find, in terms of k , the coordinates of the point that divides the interval joining $A(-2, -1)$ to $B(3, 6)$ in the ratio $k : 1 - k$. **2**

- (b) Solve for x : **3**

$$\frac{5}{x-1} \geq 1$$

- (c) If p is a positive integer, find an expression for the number of terms in the sequence **2**

$$p, p+2, p+4, \dots, 3p.$$

- (d) (i) Expand $(e^{2x} + 2)^2$. **1**

(ii) Hence evaluate $\int_0^1 (e^{2x} + 2)^2 dx$. **2**

- (e) (i) Show that $\frac{1}{x+2} - \frac{1}{x+3} = \frac{1}{(x+2)(x+3)}$. **1**

(ii) Hence find the exact value of $\int_{-1}^1 \frac{1}{(x+2)(x+3)} dx$. **2**

QUESTION TWELVE (13 marks) Use a separate writing booklet.**Marks**

- (a) A function has second derivative $y'' = 3x^3(x+3)^2(x-2)$. Determine the x -coordinates of any points of inflexion on its graph. **2**

- (b) (i) Sketch the curve $y = 8x - 4x^3$, clearly indicating the x -intercepts. (There is no need to find the stationary or inflexion points.) **2**

- (ii) Find the total area enclosed by the curve and the x -axis. **3**

- (c) A function $f(x)$ is defined by the equation $f(x) = x + \frac{4}{x}$.

- (i) Show that the function is odd. **1**

- (ii) Find $f'(x)$. **1**

- (iii) Show that the function has stationary points at $x = 2$ and $x = -2$. **1**

- (iv) Classify the two stationary points. **2**

- (v) Notice that $f(-2) = -4$ and $f(2) = 4$. So $f(-2) < f(2)$. Explain why this fact does not contradict the results in part (iii). **1**

QUESTION THIRTEEN (13 marks) Use a separate writing booklet.**Marks**

- (a) Use Simpson's rule with five function values, as well as appropriate log laws, to show

4

that $\int_1^5 \ln x \, dx \doteq \ln 57$.

- (b) A curve has gradient function
- $f'(x) = 6x^2 + px + q$
- . The curve has a stationary point at
- $(2, -4)$
- and its
- y
- intercept is 14. Find the values of
- p
- and
- q
- .

5

- (c) Suppose that the limiting sum of the series
- $v + v^2 + v^3 + \dots$
- is
- w
- .

(i) Write down a formula for w in terms of v .**1**(ii) Hence find v in terms of w .**2**

- (iii) Explain why the limiting sum of the series
- $w - w^2 + w^3 - \dots$
- is
- v
- .
-
- (You may assume that
- $|v|$
- and
- $|w|$
- are both less than one.)

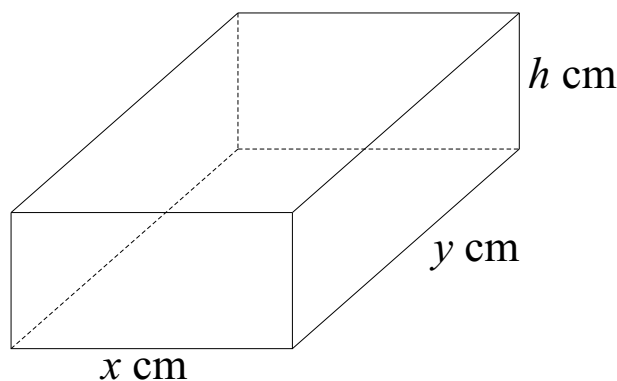
1**QUESTION FOURTEEN** (13 marks) Use a separate writing booklet.**Marks**

- (a) Solve for
- x
- :

2

$$\log_3 x + 2 = \log_3(x + 2)$$

- (b)



A closed rectangular box has dimensions x cm, y cm and h cm, as shown in the diagram above. It is to be made from 300 cm^2 of thin sheet metal, and the perimeter of its base is to be 40 cm.

- (i) Show that the volume
- V
- of the box is given by

3

$$V = 150h - 20h^2.$$

- (ii) Hence find the dimensions of the box that meets all the requirements and has the maximum possible volume.

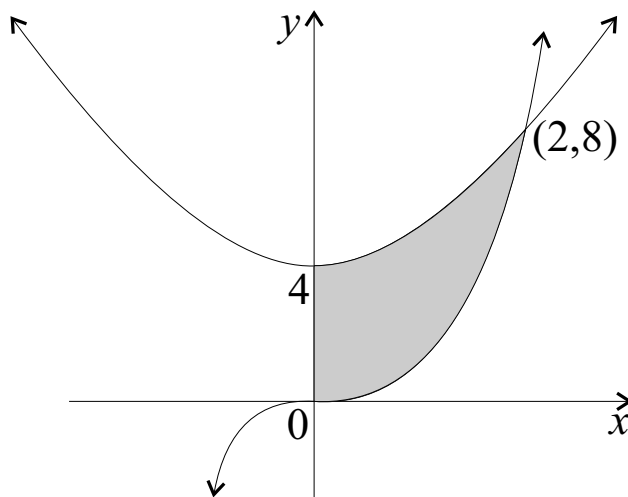
4

- (c) One root of the quadratic equation
- $ax^2 + 2bx + c = 0$
- is the reciprocal of the square of the other root.

4Prove that $a^3 + c^3 + 2abc = 0$.**Exam continues next page ...**

QUESTION FIFTEEN (13 marks) Use a separate writing booklet.**Marks**

(a)



The shaded region $\mathcal{R}$ is bounded by the curves $y = x^2 + 4$ and $y = x^3$, and the y -axis, as shown in the diagram above.

- (i) Calculate the area of $\mathcal{R}$. 3
- (ii) Determine the volume of the solid of revolution formed when $\mathcal{R}$ is rotated about the x -axis. 4
- (b) The function $y = P(x)$ is defined by $P(x) = (x - p)(x - q)(x - r)$, where p , q and r are distinct real numbers.
- (i) Sketch a possible graph of $y = P(x)$. 2
(Do NOT attempt to find the stationary or inflexion points.)
- (ii) Expand $P(x)$ and write it in the form $ax^3 + bx^2 + cx + d$. 2
- (iii) By considering the equation $P'(x) = 0$, or otherwise, prove that 2
- $$(p + q + r)^2 > 3(pq + qr + rp).$$

QUESTION SIXTEEN (13 marks) Use a separate writing booklet.

Marks

- (a) For any real number x , let $[x]$ denote the largest integer less than or equal to x .

For example $[2.9] = 2$ and $[3] = 3$.

- (i) Sketch the graph of $y = [x]$ for $0 \leq x \leq 5$.

2

- (ii) Find the value of $\int_0^5 [x] dx$.

1

- (b) Consider the function $y = \frac{\ln x}{x^n}$, where $n > 1$.

- (i) State the domain of the function.

1

- (ii) Show that there is a stationary point at $x = e^{\frac{1}{n}}$.

2

- (iii) Determine the nature of the stationary point.

2

- (iv) Sketch the graph of the function.

2

(There is no need to find the coordinates of the point of inflexion.)

- (v) Explain why $\frac{\ln x}{x^n} < \frac{1}{ne}$ for $x > e^{\frac{1}{n}}$.

1

- (vi) Deduce that $e^{\frac{1}{n-1}} > \frac{n}{n-1}$.

2

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

NAME:

CLASS: MASTER:

SYDNEY GRAMMAR SCHOOL



2015
Annual Examination
FORM V
MATHEMATICS EXTENSION 1
Monday 31st August 2015

- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

(1) n trapezia

(B)

(2) $y' = e^{-x}$

(C)

(3) $y = 1$

(A)

(4) $\frac{1}{2} \pi (2)^2 = 2\pi$

(B)

(5) $\pi \int_0^1 y dy = \frac{\pi}{2}$

(C)

(6) Decreasing and concave up.

(A)

(7) $2(x^2+1)^3$

(B)

(8) $\frac{\sin \theta}{1+\cos \theta} \cdot \frac{1-\cos \theta}{1-\cos \theta}$

$= \frac{\sin \theta (1-\cos \theta)}{\sin^2 \theta}$

(D)

(9) $(\ln e)^x = 1^x$

(C)

ONE EACH

(10)(a)(i) $-8(5-2x)^3$

(ii) $vu' + uv'$

$= e^{2x} \cdot 1 + x \cdot 2e^{2x}$

$= e^{2x}(1+2x)$

(iii) $\frac{d}{dx} \left(\frac{1}{2} \ln x \right) = \frac{1}{2x}$

(b)(i) $\int x^{-\frac{1}{2}} dx$

$= 2x^{\frac{1}{2}} + c$

$= 2\sqrt{x} + c$

(ii) $\frac{2}{3} \int \frac{3}{3x+4} dx$

$= \frac{2}{3} \ln(3x+4) + c$

(iii) $2 \int (3x+4)^{-2} dx$

$= \frac{2(3x+4)^{-1}}{-1(3)} + c$

$= \frac{-2}{3(3x+4)} + c$

(d) For continuity at $x=2$

$k(2) = 2^2 + 6$

$2k = 10$

$k = 5$

(c) $[\ln x]_e^e$

$= \ln e^3 - \ln e$

$= 3 \ln e - \ln e$

$= 2$

(11)(a) $\left(\frac{3k-2(1-k)}{k+(1-k)}, \frac{6k-(1-k)}{k+(1-k)}\right)$

$= (5k-2, 7k-1)$

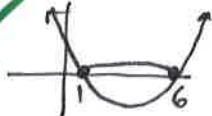
(b) Multiply both sides by $(x-1)^2$.

$5(x-1) > (x-1)^2, x \neq 1$

$(x-1)(x-1-5) \leq 0$

$(x-1)(x-6) \leq 0$

$1 < x \leq 6$



(c) Let $T_n = 3p$

then $a + (n-1)d = 3p$

$p + 2(n-1) = 3p$

$2(n-1) = 2p$

$n = p+1$

So there are $(p+1)$ terms.

(d)(i) $(e^{2x} + 2)^2 = e^{4x} + 4e^{2x} + 4$

(ii) $\int_0^1 (e^{4x} + 4e^{2x} + 4) dx$

$= \left[\frac{1}{4}e^{4x} + 2e^{2x} + 4x \right]_0^1$

$= \frac{1}{4}e^4 + 2e^2 + 4 - \left(\frac{1}{4} + 2 + 0 \right)$

$= \frac{1}{4}e^4 + 2e^2 + \frac{7}{4}$

$= \frac{1}{4}(e^4 + 8e^2 + 7)$

(e)(i) LHS = $\frac{x+3-(x+2)}{(x+2)(x+3)}$

$= \frac{1}{(x+2)(x+3)}$

$= \text{RHS}$

(ii) $\int_{-1}^1 \left(\frac{1}{x+2} - \frac{1}{x+3} \right) dx$

$= [\ln(x+2) - \ln(x+3)]_{-1}^1$

$= \left[\ln\left(\frac{x+2}{x+3}\right) \right]_{-1}^1$

$= \ln\frac{3}{4} - \ln\frac{1}{2}$

$= \ln\frac{3}{2}$

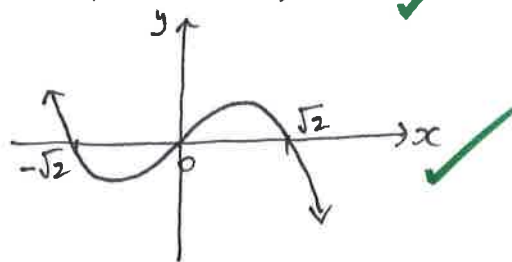
(12)(a)

x	-4	-3	-1	0	1	2	3
y''	+	0	+	0	-	0	+

The concavity changes at $x=0$ and $x=2$. (But NOT at $x=-3$.)

(b)(i) $y = 4x(2-x^2)$
 $= 4x(\sqrt{2}+x)(\sqrt{2}-x)$

x -intercepts at $0, \pm\sqrt{2}$.



(ii) $A = 2 \int_0^{\sqrt{2}} (8x - 4x^3) dx$

$= 2 [4x^2 - x^4]_0^{\sqrt{2}}$

$= 2(8-4)$

$= 8$

(c)(i) $f(-x) = -x + \frac{4}{-x} = -(x + \frac{4}{x}) = -f(x)$

(ii) $f(x) = x + \frac{4}{x}$

so $f'(x) = 1 - 4x^{-2}$

$= 1 - \frac{4}{x^2}$

(iii) Let $f'(x) = 0$ for stationary points.

Then $1 - \frac{4}{x^2} = 0$

$x^2 = 4$

$x = \pm 2$.

(iv) $f''(x) = 8x^{-3}$

so $f''(-2) < 0$ and $f''(2) > 0$.

So there is a maximum turning point at $x=-2$ and a minimum turning point at $x=2$.

(v) $(-2, -4)$ and $(2, 4)$ lie on different branches of the curve, because there is a vertical asymptote at $x=0$.

(13)(a)

x	1	2	3	4	5
$y = \ln x$	0	$\ln 2$	$\ln 3$	$\ln 4$	$\ln 5$
	(y_0)	(y_1)	(y_2)	(y_3)	(y_4)

$$\int_1^5 \ln x \, dx \doteq \frac{h}{3} (y_0 + y_4 + 4(y_1 + y_3) + 2y_2)$$

$$= \frac{1}{3} (\ln 5 + 4(\ln 2 + \ln 4) + 2\ln 3)$$

$$= \frac{1}{3} \ln (5 \times 2^4 \times 4^4 \times 3^2)$$

$$= \ln 184320^{\frac{1}{3}}$$

$$= \ln 56.91 \dots$$

$$\doteq \ln 57$$

(b) $f'(2) = 0$ so $24 + 2p + q = 0$ (1)

$$f(x) = 2x^3 + \frac{1}{2}px^2 + qx + c$$

$$f(0) = 14$$
 so $c = 14$

$$f(2) = -4$$
 so $16 + 2p + 2q + 14 = -4$

$$\text{so } 17 + p + q = 0$$
 (2)

(1) - (2): $7 + p = 0$

$$\text{so } p = -7 \text{ and } q = -10.$$

(c)(i) We are given that $\frac{v}{1-v} = w$.

(ii) So $v = w - wv$

$$v + wv = w$$

$$v(1+w) = w$$

$$v = \frac{w}{1+w}$$

(iii) $v = \frac{w}{1-(-w)}$

$$= w - w^2 + w^3 - \dots$$

(assuming that both series converge)

(14)(a) $2 = \log_3(x+2) - \log_3 x$

$$\log_3 \frac{x+2}{x} = 2$$

$$\frac{x+2}{x} = 9$$

$$x+2 = 9x$$

$$x = \frac{1}{4}$$

(b)(i) Given $2x + 2y = 40$

$$\text{so } x + y = 20$$
 (1)

and given $2xy + 2xh + 2yh = 300$

$$\text{so } xy + h(x+y) = 150$$
 (using (1))
$$\text{so } xy + 20h = 150$$

$$\text{so } xy = 150 - 20h$$
 (2)

$$V = xyh$$

$$= (150 - 20h)h \text{ (using (2))}$$

$$= 150h - 20h^2$$

(ii) $\frac{dV}{dh} = 150 - 40h$

Let $\frac{dV}{dh} = 0$ for stationary points.

$$\text{Then } h = \frac{15}{4}$$

$$\frac{d^2V}{dh^2} = -40 < 0,$$

so V is maximised when $h = \frac{15}{4}$.

When $h = \frac{15}{4}$,

$$xy = 75 \text{ and } x + y = 20.$$

By inspection, $x = 15$ and $y = 5$ (or vice versa),

so the box has base 15cm by 5cm and its height is 3.75cm.

(c) Let the roots be α and $\frac{1}{\alpha^2}$.

$$\text{Then } \alpha + \frac{1}{\alpha^2} = -\frac{2b}{a}$$
 (1)
$$\text{and } \alpha \cdot \frac{1}{\alpha^2} = \frac{c}{a}$$

$$\text{so } \alpha = \frac{a}{c}$$
 (2)

Substituting (2) into (1):

$$\frac{a}{c} + \frac{c^2}{a^2} = -\frac{2b}{a}$$

Multiply both sides by a^2c :

$$a^3 + c^3 = -2bac$$

$$\text{so } a^3 + c^3 + 2abc = 0.$$

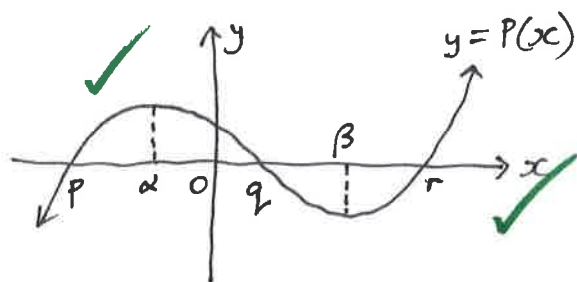
(15)(a)(i)

$$\begin{aligned}
 \text{Area} &= \int_0^2 (x^2 + 4 - x^3) dx \\
 &= \left[\frac{1}{3}x^3 + 4x - \frac{1}{4}x^4 \right]_0^2 \\
 &= \frac{8}{3} + 8 - 4 \\
 &= \frac{20}{3} u^2
 \end{aligned}$$

(ii)

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^2 ((x^2 + 4)^2 - (x^3)^2) dx \\
 &= \pi \int_0^2 (x^4 + 8x^2 + 16 - x^6) dx \\
 &= \pi \left[\frac{1}{5}x^5 + \frac{8}{3}x^3 + 16x - \frac{1}{7}x^7 \right]_0^2 \\
 &= \pi \left(\frac{32}{5} + \frac{64}{3} + 32 - \frac{128}{7} \right) \\
 &= \frac{4352\pi}{105} u^3
 \end{aligned}$$

(b)(i)



(ii) Expanding, we have

$$P(x) = x^3 - (p+q+r)x^2 + (pq+qr+rp)x - pqr$$

$$\text{so } P'(x) = 3x^2 - 2(p+q+r)x + (pq+qr+rp)$$

The equation $P'(x) = 0$ has two distinct real roots (α and β in the diagram above),

$$\text{so } \Delta > 0$$

$$\text{so } 4(p+q+r)^2 - 12(pq+qr+rp) > 0$$

$$\text{so } (p+q+r)^2 > 3(pq+qr+rp)$$

(ii) (OTHERWISE)

$$(p-q)^2 > 0$$

$$\text{so } p^2 + q^2 > 2pq \quad (1)$$

$$\text{Similarly } q^2 + r^2 > 2qr \quad (2)$$

$$\text{and } r^2 + p^2 > 2rp \quad (3)$$

$$(1) + (2) + (3) :$$

$$2p^2 + 2q^2 + 2r^2 > 2pq + 2qr + 2rp$$

$$\text{so } p^2 + q^2 + r^2 > pq + qr + rp \quad (4)$$

Now,

$$(p+q+r)^2 = p^2 + q^2 + r^2 + 2(pq+qr+rp)$$

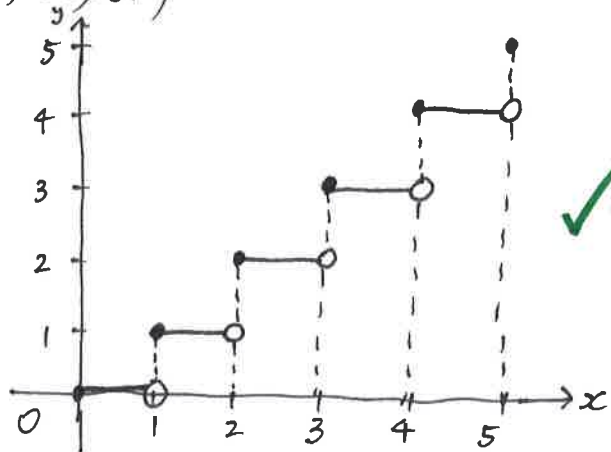
From (4), it follows that

$$(p+q+r)^2 > pq+qr+rp + 2(pq+qr+rp)$$

so

$$(p+q+r)^2 > 3(pq+qr+rp).$$

(16)(a)(i)



$$(ii) \int_0^5 [x] dx = 0 + 1 + 2 + 3 + 4 = 10$$

(b)(i) Domain is $x > 0$.

$$(ii) y' = \frac{x^n \cdot \frac{1}{x} - nx^{n-1} \cdot \ln x}{(x^n)^2}$$

$$= \frac{x^{n-1}(1 - n \ln x)}{x^{2n}}$$

Let $y' = 0$ for stationary points.

$x > 0$, so the only solution occurs when $\ln x = \frac{1}{n}$
i.e. when $x = e^{\frac{1}{n}}$.

(iii)	x	$e^{\frac{1}{2n}}$	$e^{\frac{1}{n}}$	$e^{\frac{2}{n}}$
	y'	$\frac{e^{\frac{n-1}{2n}}(1-\frac{1}{2})}{e}$	0	$\frac{e^{\frac{2n-2}{n}}(1-2)}{e^4}$

There is a maximum turning point at $x = e^{\frac{1}{n}}$.

(some working or explanation required)

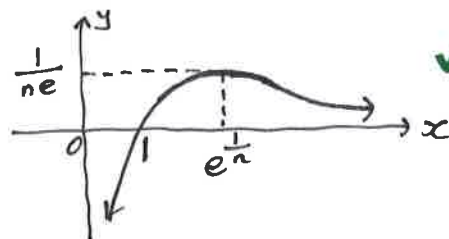
(5)

(iv) x-intercept at $(1, 0)$.

$$\text{When } x = e^{\frac{1}{n}}, \\ y = \frac{1}{n} \div (e^{\frac{1}{n}})^n \\ = \frac{1}{ne}$$

As $x \rightarrow \infty$, $y \rightarrow 0^+$.

As $x \rightarrow 0^+$, $y \rightarrow -\infty$.



(v) From the graph, when $x > e^{\frac{1}{n}}$, $y < \frac{1}{ne}$,
so when $x > e^{\frac{1}{n}}$, $\frac{\ln x}{x^n} < \frac{1}{ne}$.
(since the max y-value is $\frac{1}{ne}$)

(vi) Let $x = e^{\frac{1}{n-1}}$ ($> e^{\frac{1}{n}}$),
then $\frac{1}{n-1} \div e^{\frac{n}{n-1}-1} < \frac{1}{ne}$
so $\frac{n}{n-1} < e^{\frac{n}{n-1}-1}$
so $e^{\frac{1}{n-1}} > \frac{n}{n-1}$.